

Cyclepoint with Schaefer 2018 example

2025-02-12

```
library(tidyverse)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.2
```

```
library(lme4)
library(splines)
library(splines2)
library(easystats)
library(ggeffects)
library(merTools)
library(mgcv)
library(gamm4)
library(sjPlot)
library(patchwork)
##### Read file #####
MRIGam <- read.csv("parcel_activation.csv")
cyc <- paste('cyclepoint ', toString(seq(0.0,1.0,by=0.01)), ' ')

parclist <- c(15, 65, 73, 14, 51, 71, 78, 9, 57, 76,
              8, 6, 77, 55, 21, 60, 89, 99, 58, 23,
              53, 3, 67, 36, 49, 30, 56)
```

Learning Block

```
##### zstat4 #####
for (i in parclist){
  var = paste0("parcel_", i)
  formula = as.formula(paste(var, "s(cyclepoint,bs='cc')", sep = " ~ "))
  wholemg = gamm4(formula,data = MRIGam)
  mod = summary(wholemg$gam)
  print(mod)

  plot = plot(ggpredict(wholemg$gam, terms=c(cyc))) +
    #geom_line(color='#35b779') +
    geom_ribbon(aes(ymin = conf.low, ymax = conf.high),fill='#943ddc', alpha = 0.15) +
    labs(x= 'cyclepoint',y='activation') +
    #annotate("text", x = 0.5, y = 3.2, label = var) +
    theme_classic()+
    theme(axis.line=element_line(size=0.6),
          axis.text=element_text(size=10),
          axis.title=element_text(size=12),
```

```

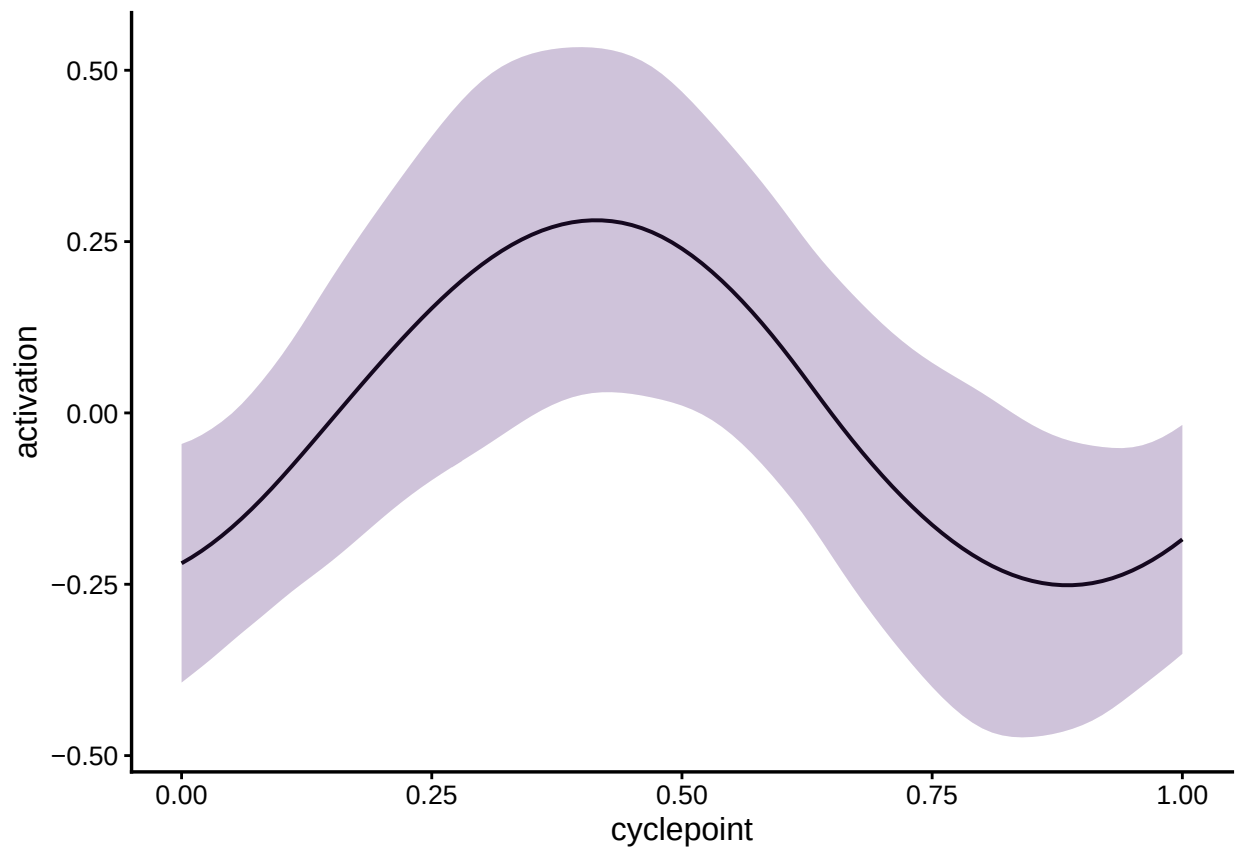
    plot.title = element_blank(),
    legend.title = element_blank(),
    legend.position='top')
  print(plot)
}

```

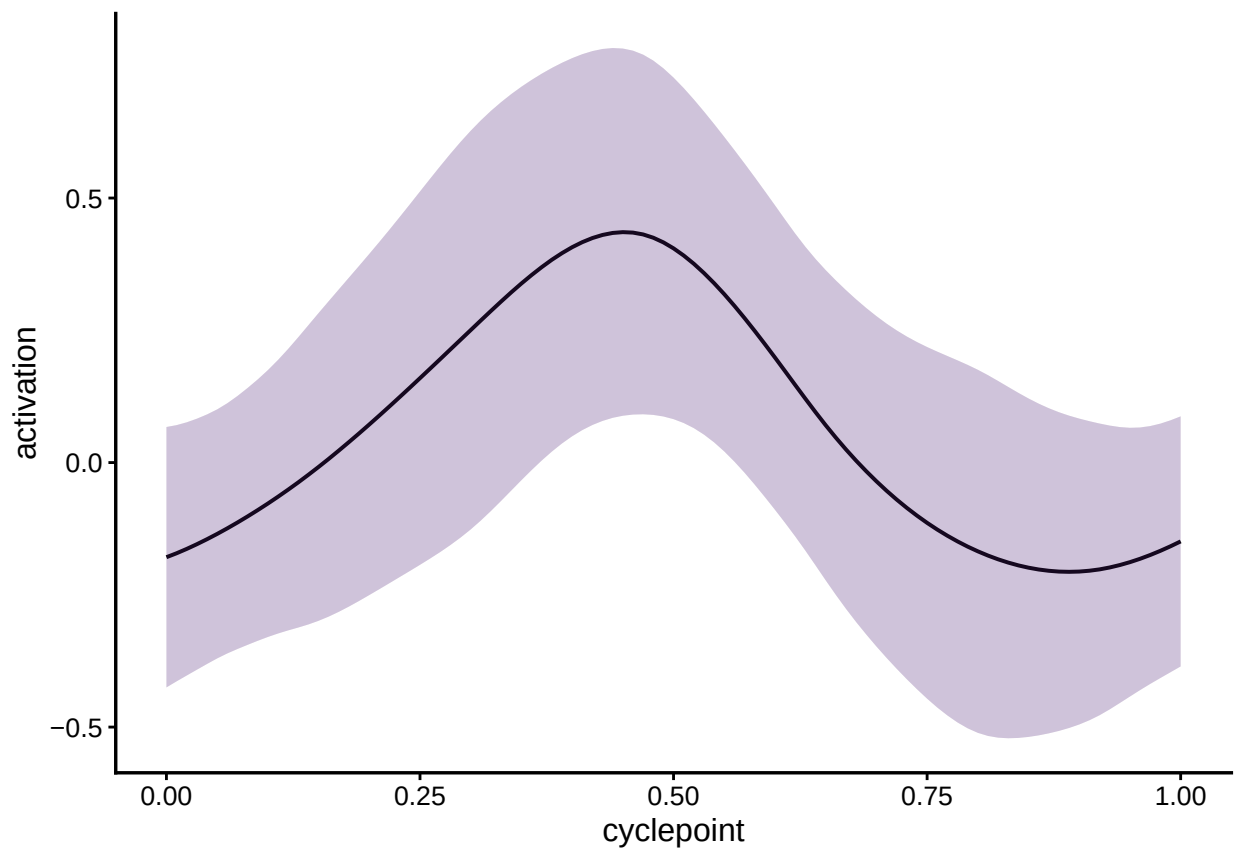
```

##
## Family: gaussian
## Link function: identity
##
## Formula:
## parcel_15 ~ s(cyclepoint, bs = "cc")
##
## Parametric coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.03871    0.05687  -0.681    0.5
##
## Approximate significance of smooth terms:
##              edf Ref.df      F p-value
## s(cyclepoint) 2.168      8 8.433 0.00401 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.212
## lmer.REML = 42.341  Scale est. = 0.13581  n = 42

```

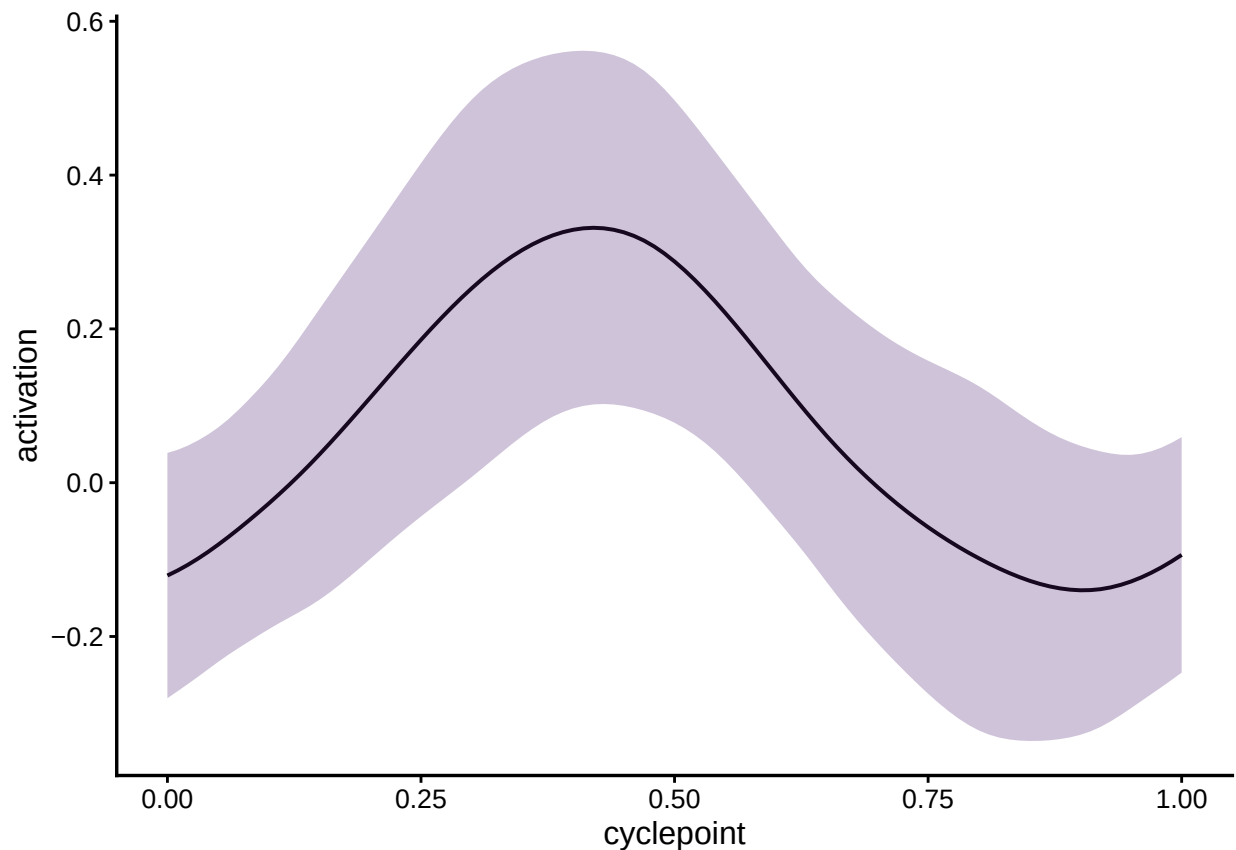


```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## parcel_65 ~ s(cyclepoint, bs = "cc")
##
## Parametric coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.01978    0.08090   0.244   0.808
##
## Approximate significance of smooth terms:
##             edf Ref.df    F p-value
## s(cyclepoint) 2.117     8 3.106  0.0152 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.169
## lmer.REML = 71.102  Scale est. = 0.27491  n = 42
```



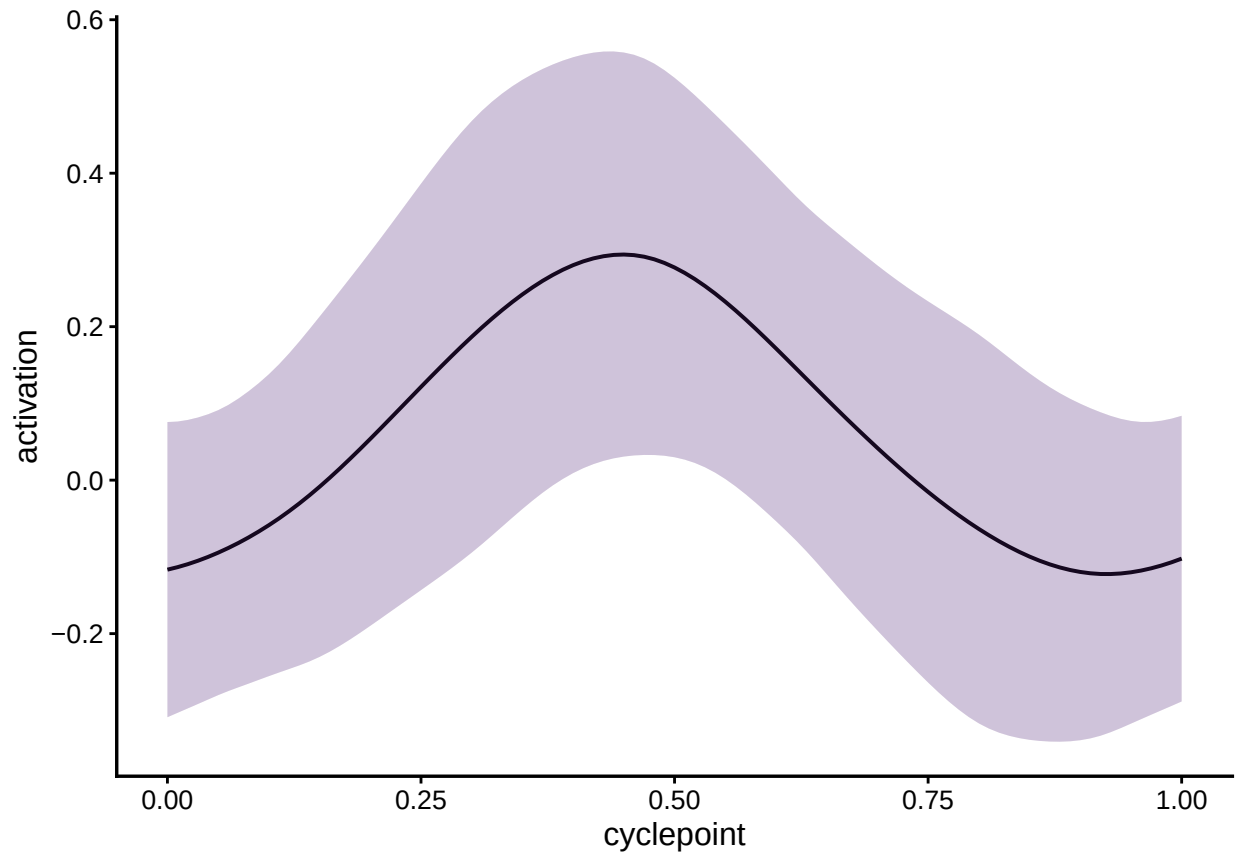
```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## parcel_73 ~ s(cyclepoint, bs = "cc")
```

```
##
## Parametric coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.03256    0.05216   0.624   0.536
##
## Approximate significance of smooth terms:
##           edf Ref.df      F p-value
## s(cyclepoint) 2.159      8 9.029 0.00624 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.199
## lmer.REML = 35.234  Scale est. = 0.11427  n = 42
```



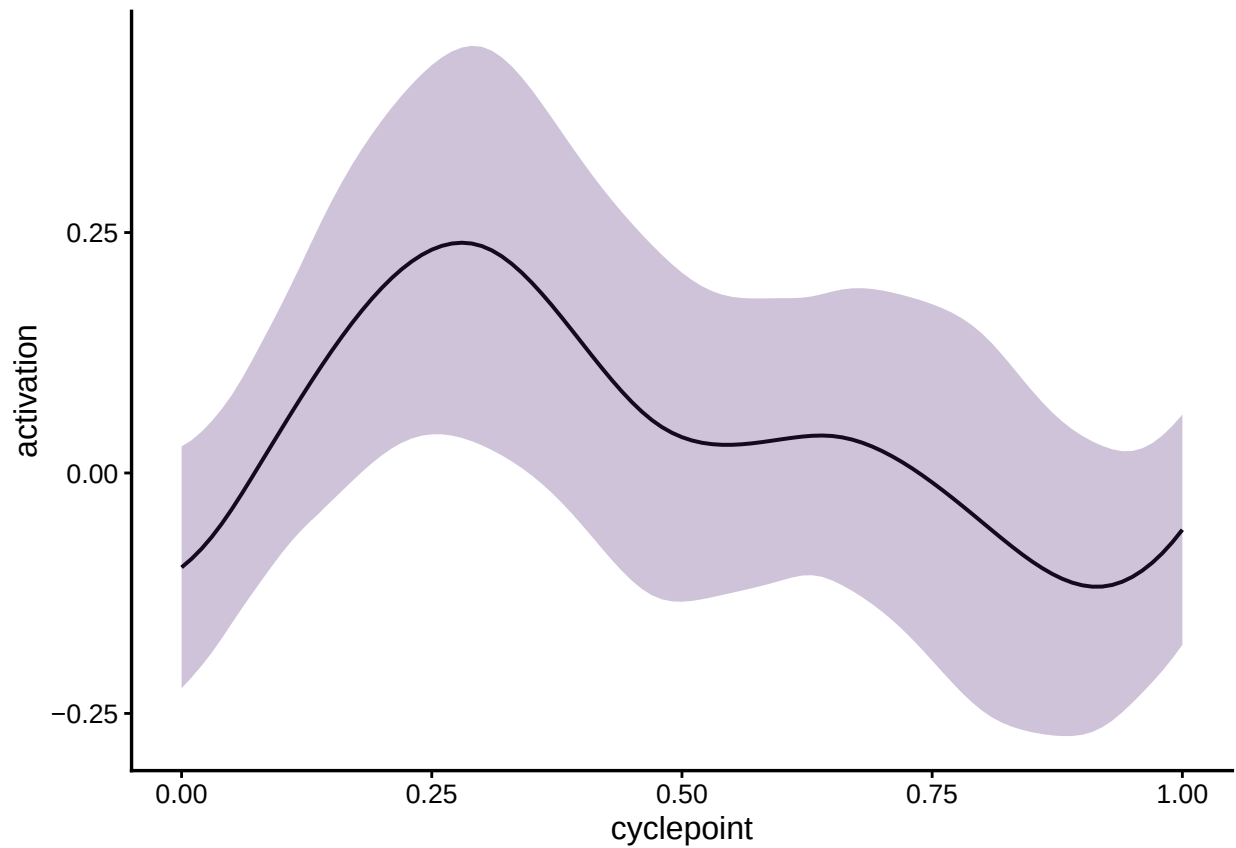
```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## parcel_14 ~ s(cyclepoint, bs = "cc")
##
## Parametric coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.02947    0.06612   0.446   0.658
##
## Approximate significance of smooth terms:
```

```
##               edf Ref.df    F p-value
## s(cyclepoint) 1.761      8 3.34 0.0219 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) = 0.134
## lmer.REML = 53.562  Scale est. = 0.1836    n = 42
```

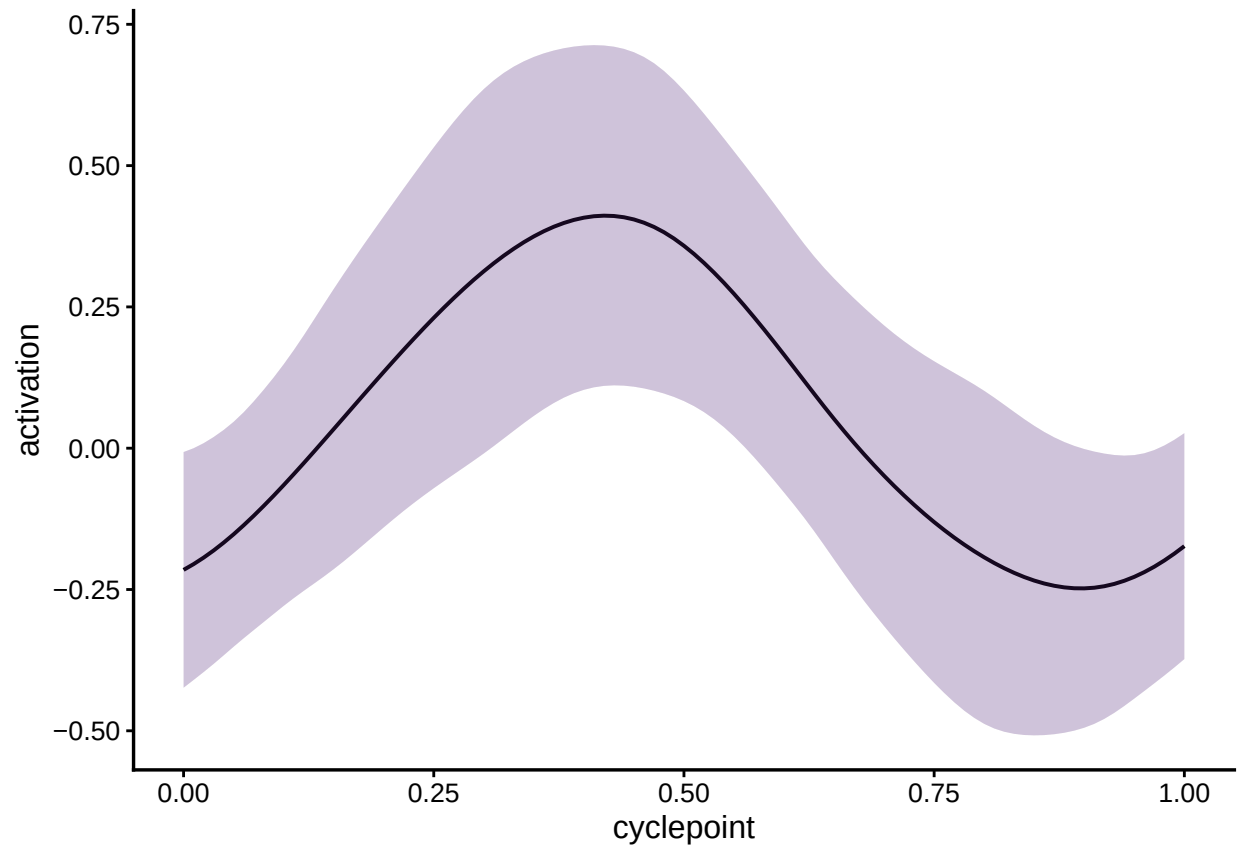


```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## parcel_51 ~ s(cyclepoint, bs = "cc")
##
## Parametric coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.01726   0.03811    0.453   0.653
##
## Approximate significance of smooth terms:
##               edf Ref.df    F p-value
## s(cyclepoint) 2.873      8 12.2 0.0504 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) = 0.174
```

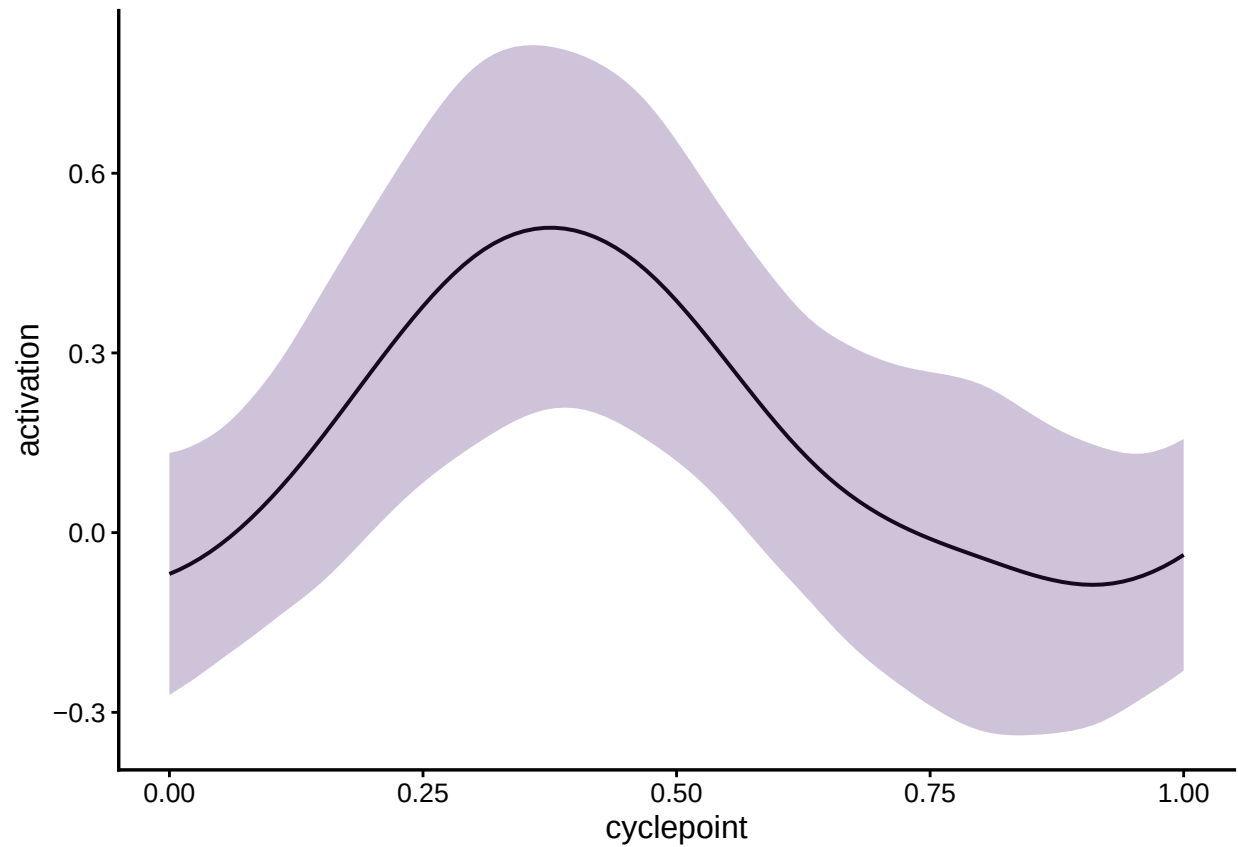
```
## lmer.REML = 11.753  Scale est. = 0.060989  n = 42
```



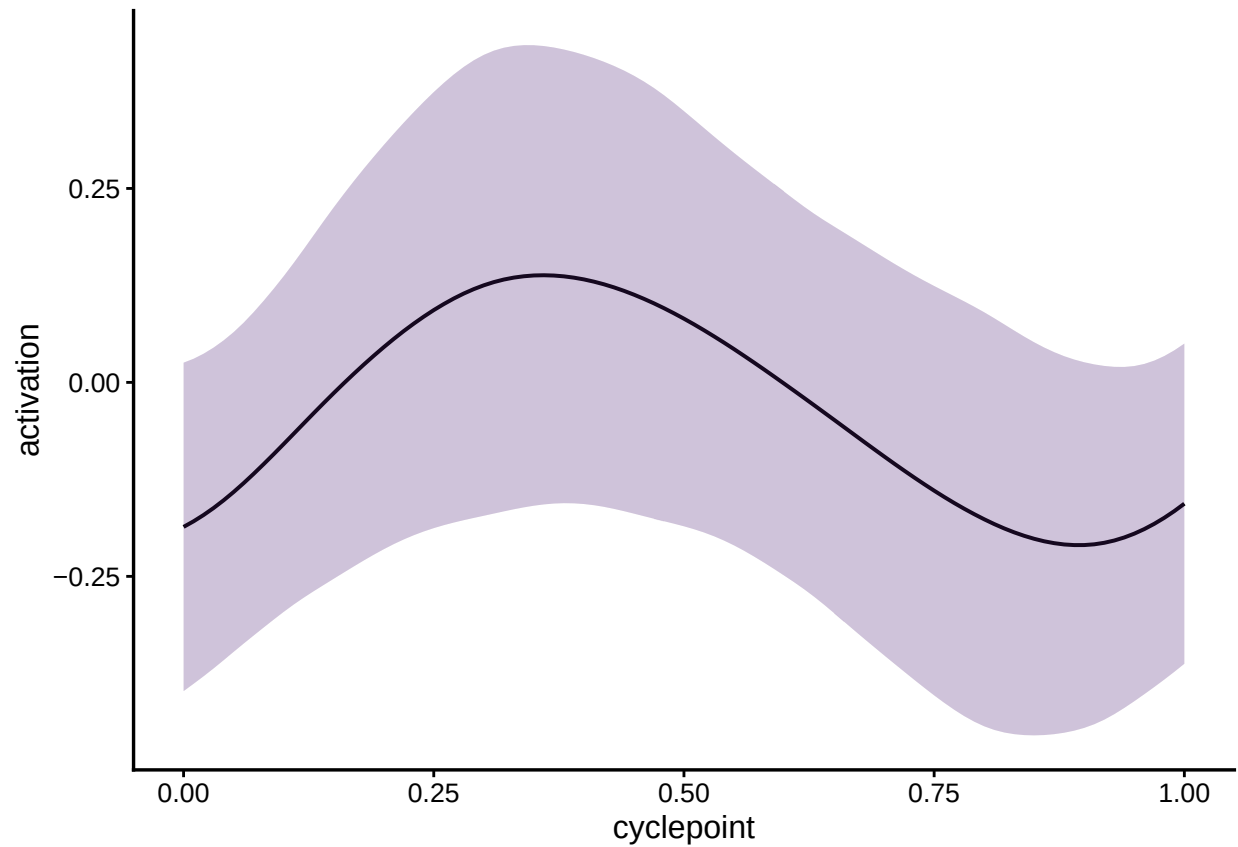
```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## parcel_71 ~ s(cyclepoint, bs = "cc")
##
## Parametric coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.007014   0.067831   0.103   0.918
##
## Approximate significance of smooth terms:
##               edf Ref.df      F p-value
## s(cyclepoint)  2.204     8 6.356  0.0033 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.221
## lmer.REML = 56.909  Scale est. = 0.19324  n = 42
```



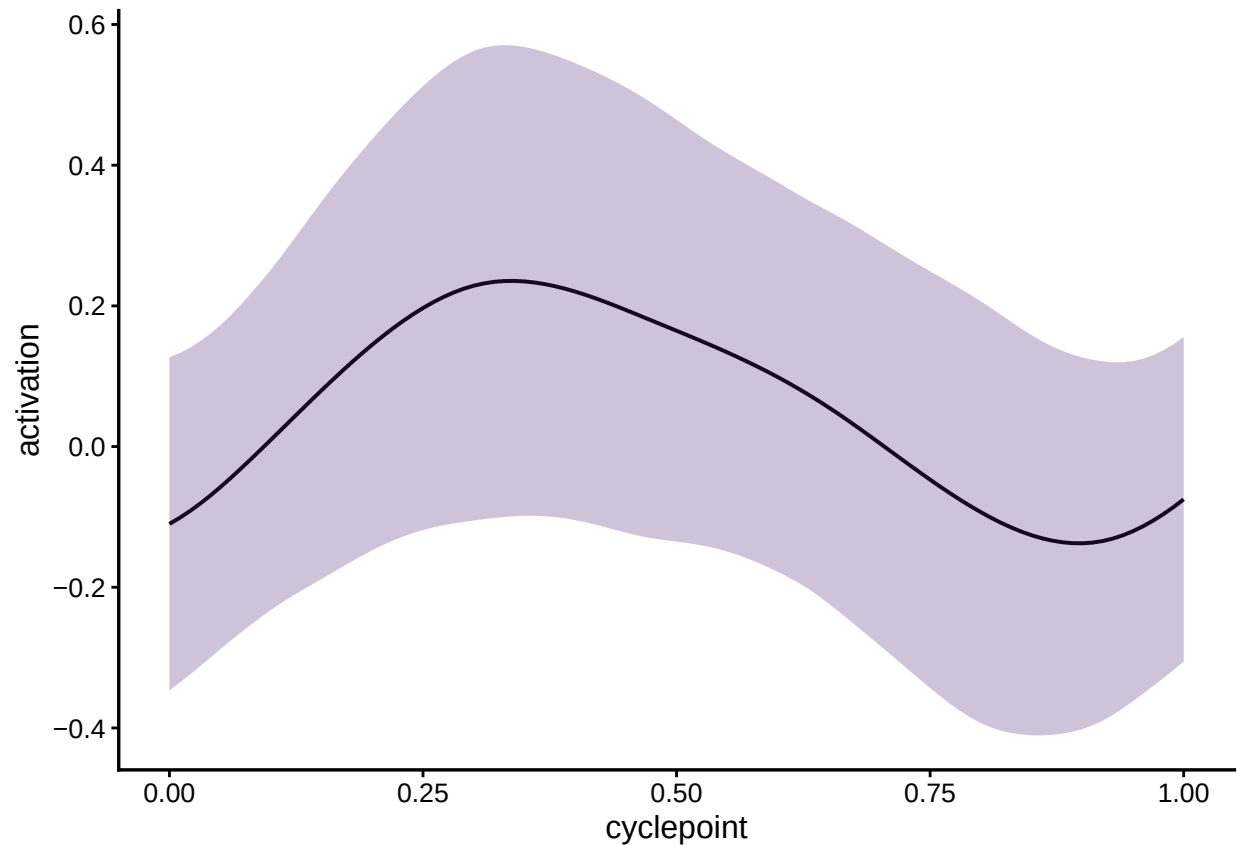
```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## parcel_78 ~ s(cyclepoint, bs = "cc")
##
## Parametric coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.10971    0.06508   1.686   0.0999 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##              edf Ref.df      F p-value
## s(cyclepoint) 2.287     8 5.635 0.00962 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.194
## lmer.REML = 53.765  Scale est. = 0.17789  n = 42
```



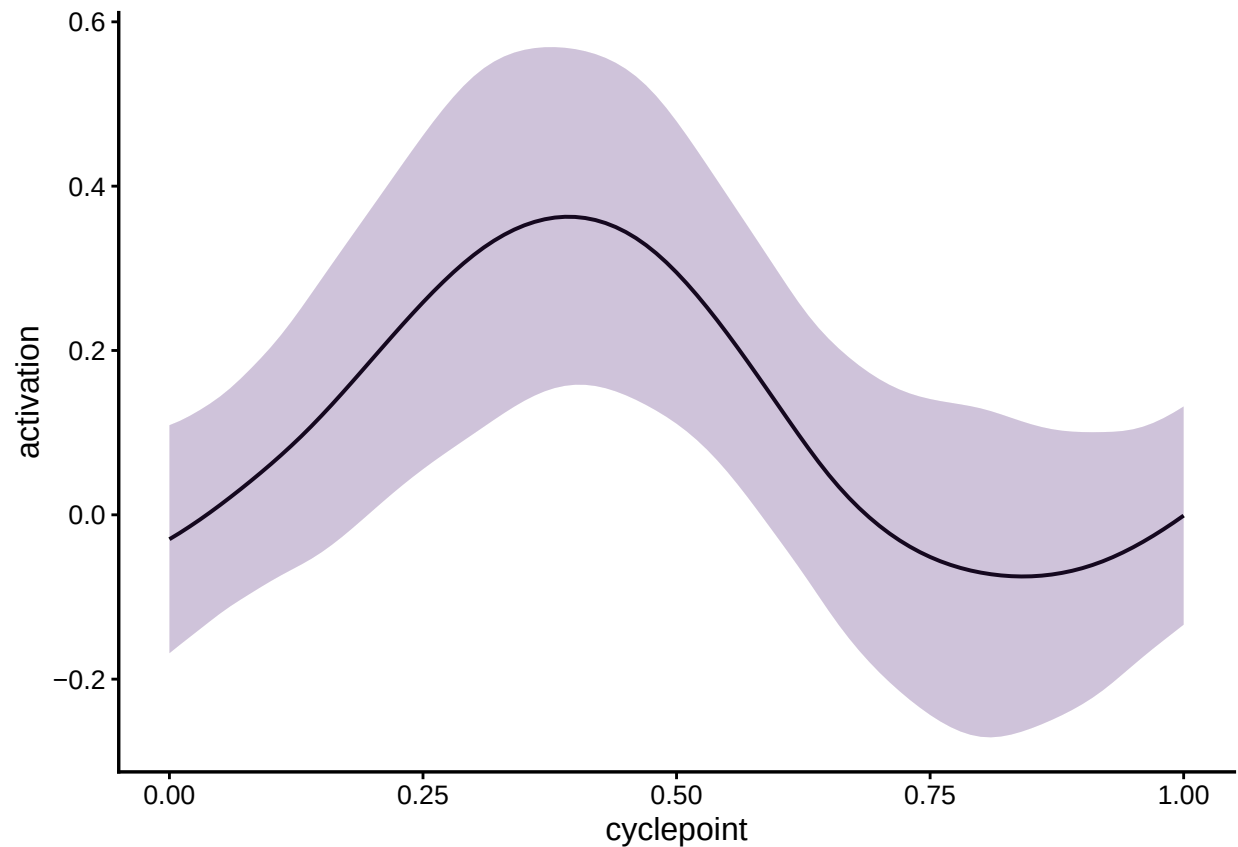
```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## parcel_9 ~ s(cyclepoint, bs = "cc")
##
## Parametric coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.06487    0.07566  -0.857   0.396
##
## Approximate significance of smooth terms:
##             edf Ref.df    F p-value
## s(cyclepoint) 1.486     8 1.211   0.101
##
## R-sq.(adj) =  0.0778
## lmer.REML = 63.951  Scale est. = 0.24044  n = 42
```

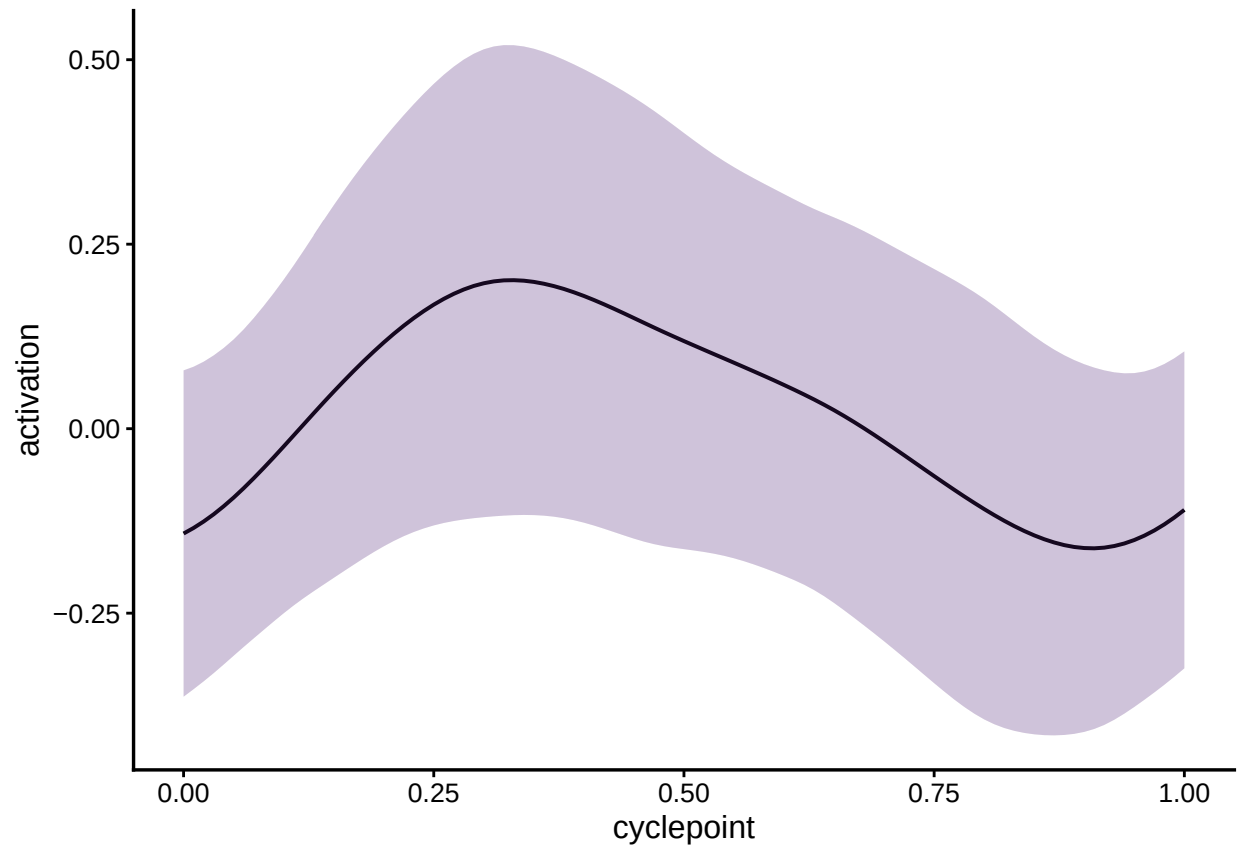
```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## parcel_57 ~ s(cyclepoint, bs = "cc")
##
## Parametric coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.02379    0.08425   0.282   0.779
##
## Approximate significance of smooth terms:
##             edf Ref.df    F p-value
## s(cyclepoint) 1.515     8 0.94  0.124
##
## R-sq.(adj) =  0.0746
## lmer.REML = 72.83  Scale est. = 0.2981    n = 42
```



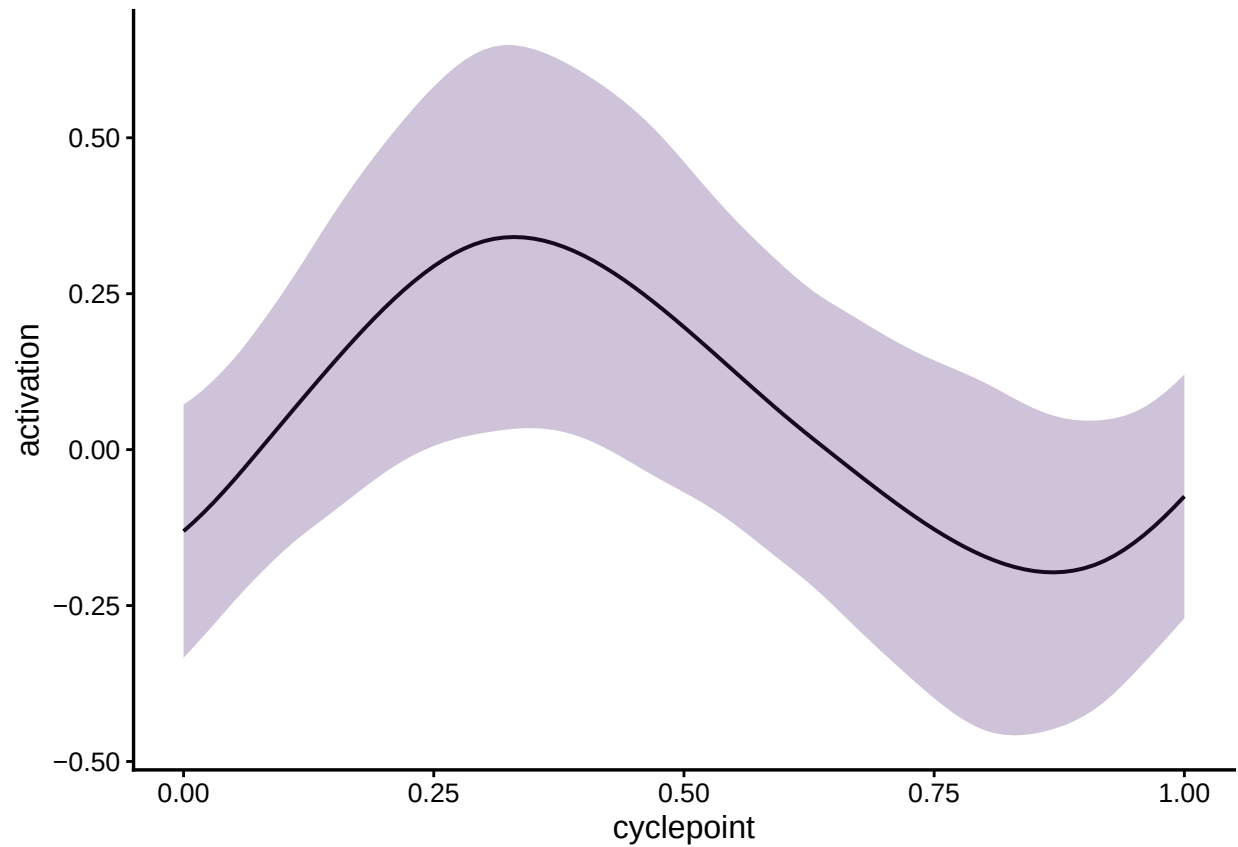
```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## parcel_76 ~ s(cyclepoint, bs = "cc")
##
## Parametric coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.08424    0.04452   1.892   0.066 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##               edf Ref.df    F p-value
## s(cyclepoint) 2.327     8 12.62 0.00759 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.205
## lmer.REML = 22.763  Scale est. = 0.083263  n = 42
```



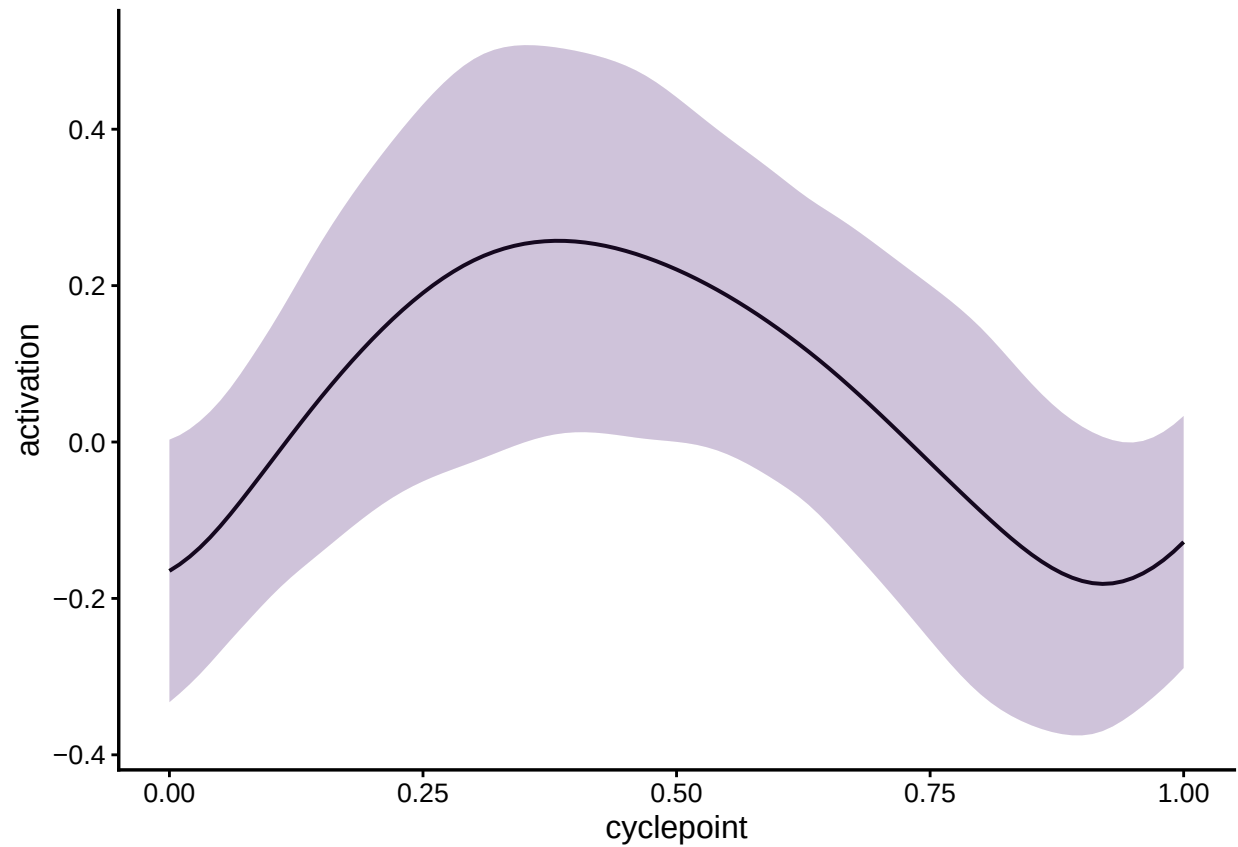
```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## parcel_8 ~ s(cyclepoint, bs = "cc")
##
## Parametric coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.009305   0.077436  -0.12    0.905
##
## Approximate significance of smooth terms:
##               edf Ref.df    F p-value
## s(cyclepoint) 1.625     8 1.19  0.124
##
## R-sq.(adj) =  0.081
## lmer.REML = 66.18  Scale est. = 0.25185  n = 42
```



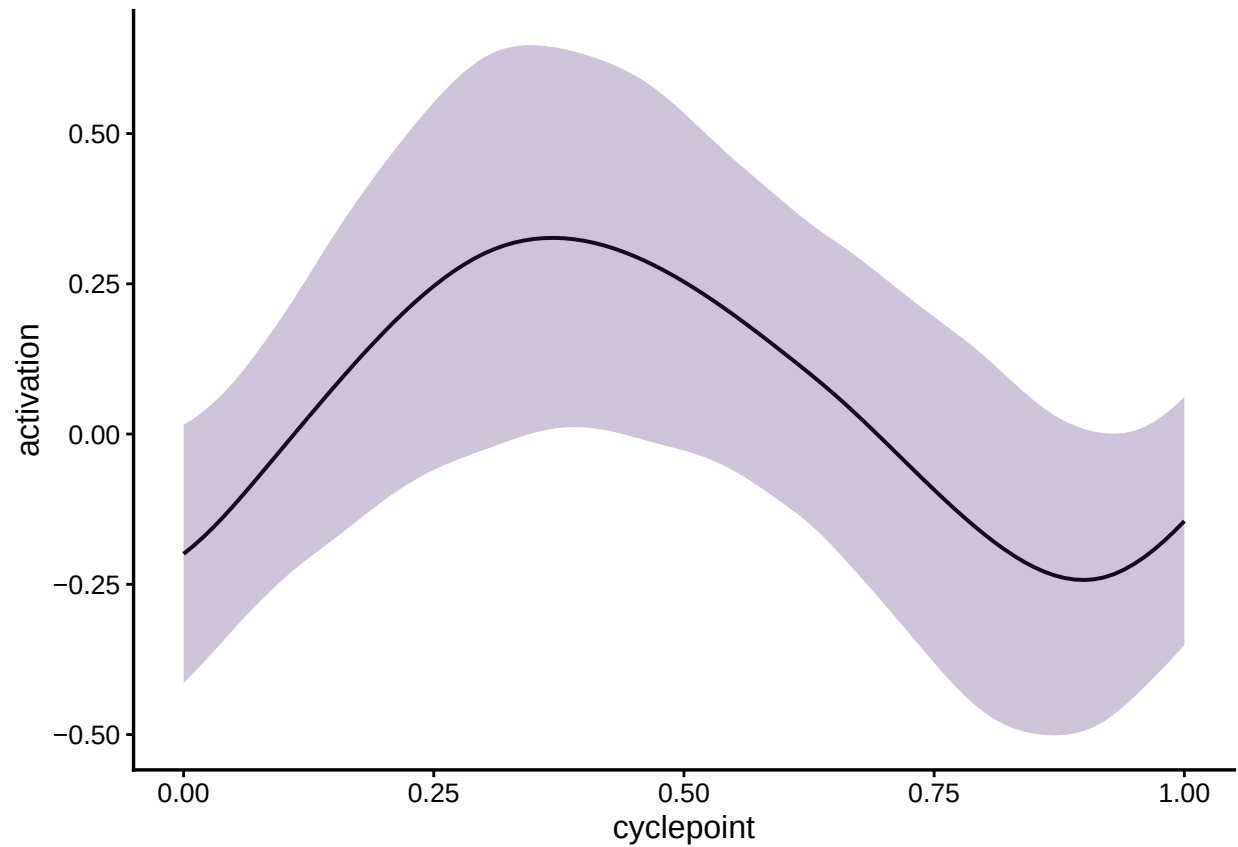
```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## parcel_6 ~ s(cyclepoint, bs = "cc")
##
## Parametric coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.02309    0.06753   0.342   0.734
##
## Approximate significance of smooth terms:
##               edf Ref.df      F p-value
## s(cyclepoint) 2.019     8 3.474 0.0284 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.145
## lmer.REML = 56.001  Scale est. = 0.19154  n = 42
```



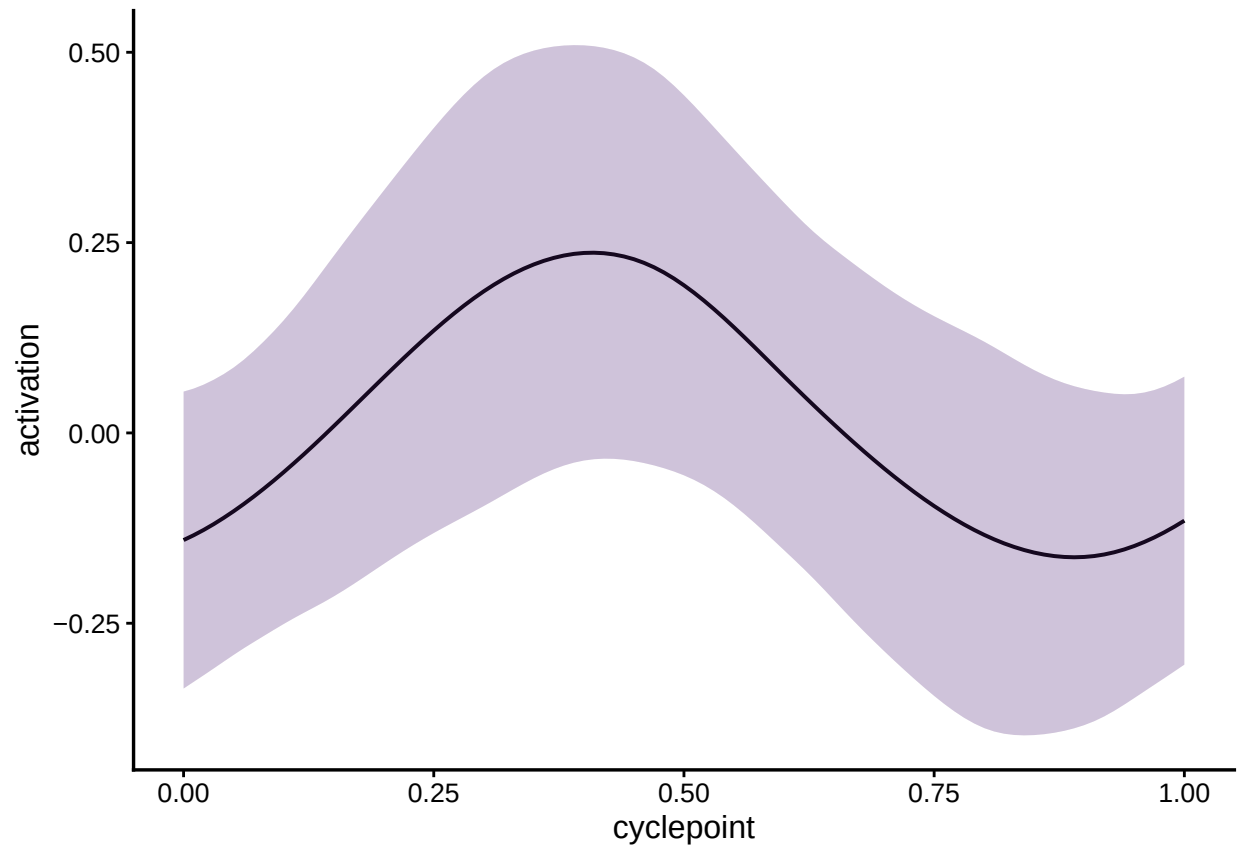
```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## parcel_77 ~ s(cyclepoint, bs = "cc")
##
## Parametric coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.01751    0.05510   0.318   0.752
##
## Approximate significance of smooth terms:
##             edf Ref.df    F p-value
## s(cyclepoint) 2.129     8 7.165  0.0101 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.183
## lmer.REML = 39.638  Scale est. = 0.12751  n = 42
```



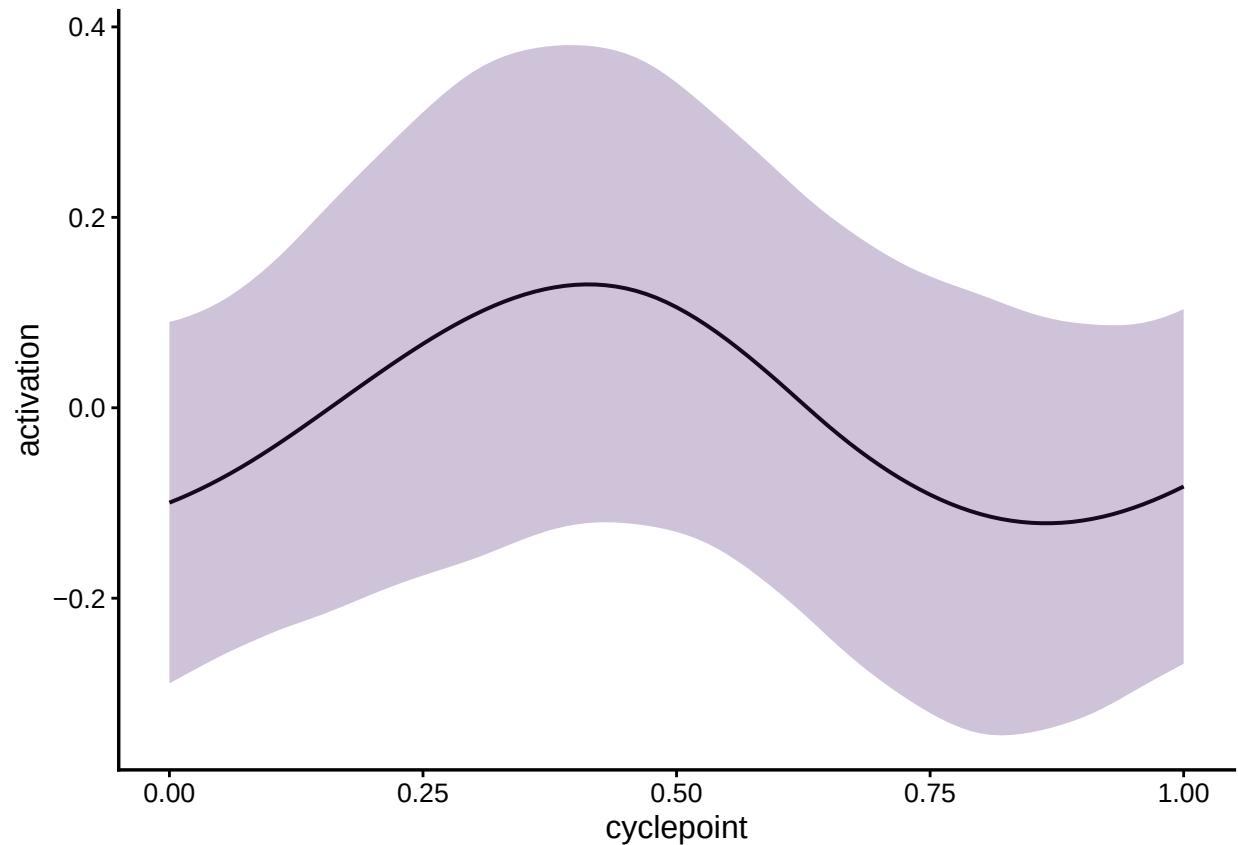
```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## parcel_55 ~ s(cyclepoint, bs = "cc")
##
## Parametric coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.009461  0.071061   0.133   0.895
##
## Approximate significance of smooth terms:
##             edf Ref.df    F p-value
## s(cyclepoint) 2.058     8 3.88 0.0146 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.167
## lmer.REML = 60.292  Scale est. = 0.21208   n = 42
```



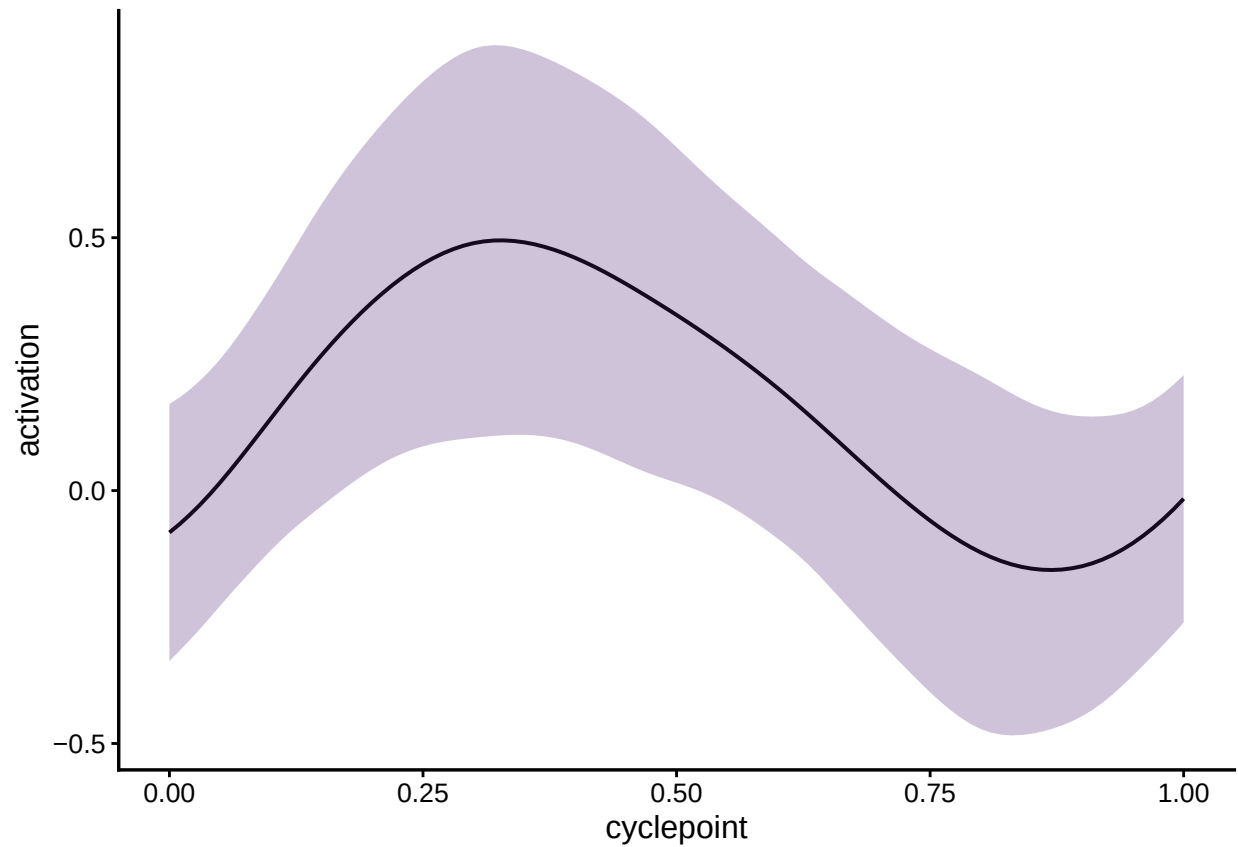
```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## parcel_21 ~ s(cyclepoint, bs = "cc")
##
## Parametric coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.01041    0.06775  -0.154   0.879
##
## Approximate significance of smooth terms:
##              edf Ref.df      F p-value
## s(cyclepoint) 1.684     8 2.382 0.0462 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.109
## lmer.REML = 55.369  Scale est. = 0.19278  n = 42
```



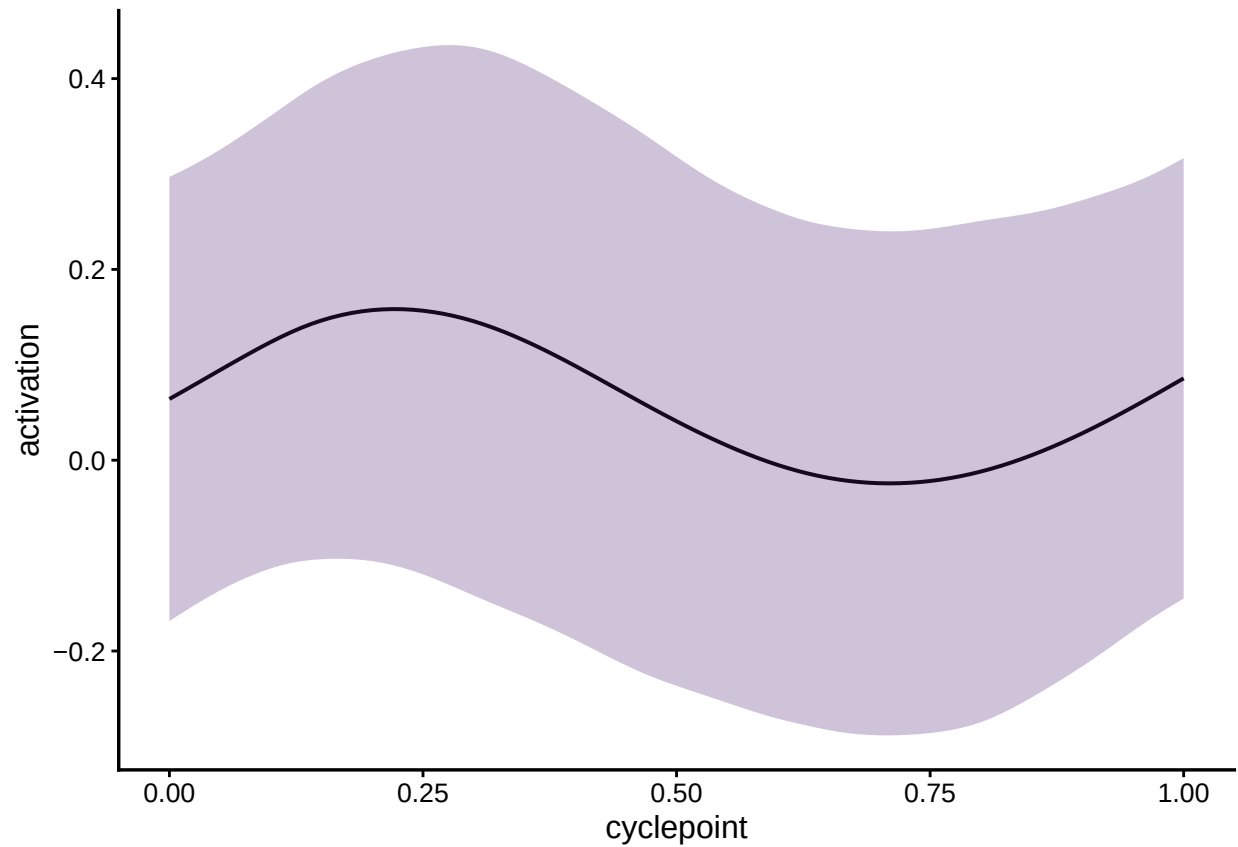
```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## parcel_60 ~ s(cyclepoint, bs = "cc")
##
## Parametric coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.02289    0.07039  -0.325   0.747
##
## Approximate significance of smooth terms:
##             edf Ref.df    F p-value
## s(cyclepoint) 1.251     8 0.862  0.156
##
## R-sq.(adj) =  0.0558
## lmer.REML = 57.524  Scale est. = 0.20809  n = 42
```

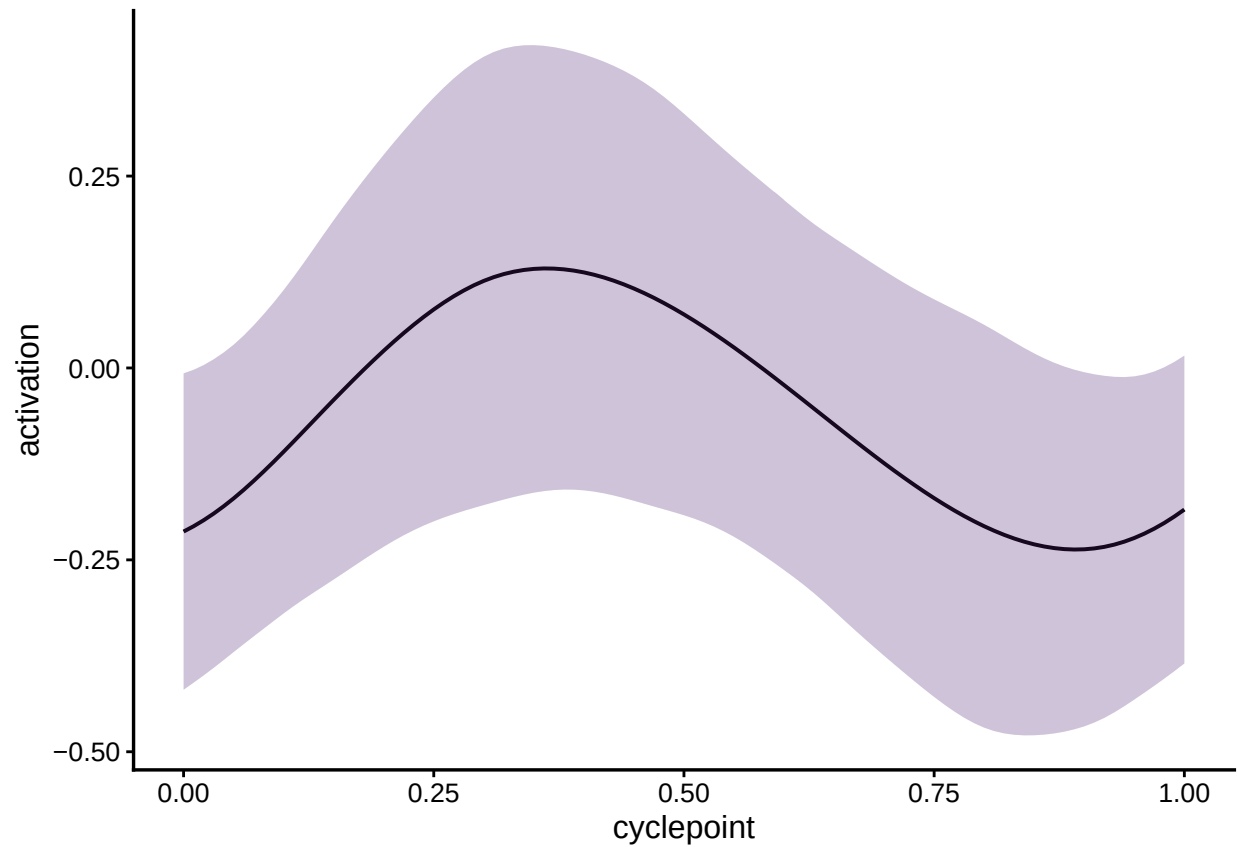
```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## parcel_89 ~ s(cyclepoint, bs = "cc")
##
## Parametric coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.12360    0.08453   1.462   0.152
##
## Approximate significance of smooth terms:
##              edf Ref.df    F p-value
## s(cyclepoint) 2.021     8 2.314 0.0278 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.145
## lmer.REML = 74.415  Scale est. = 0.30008  n = 42
```



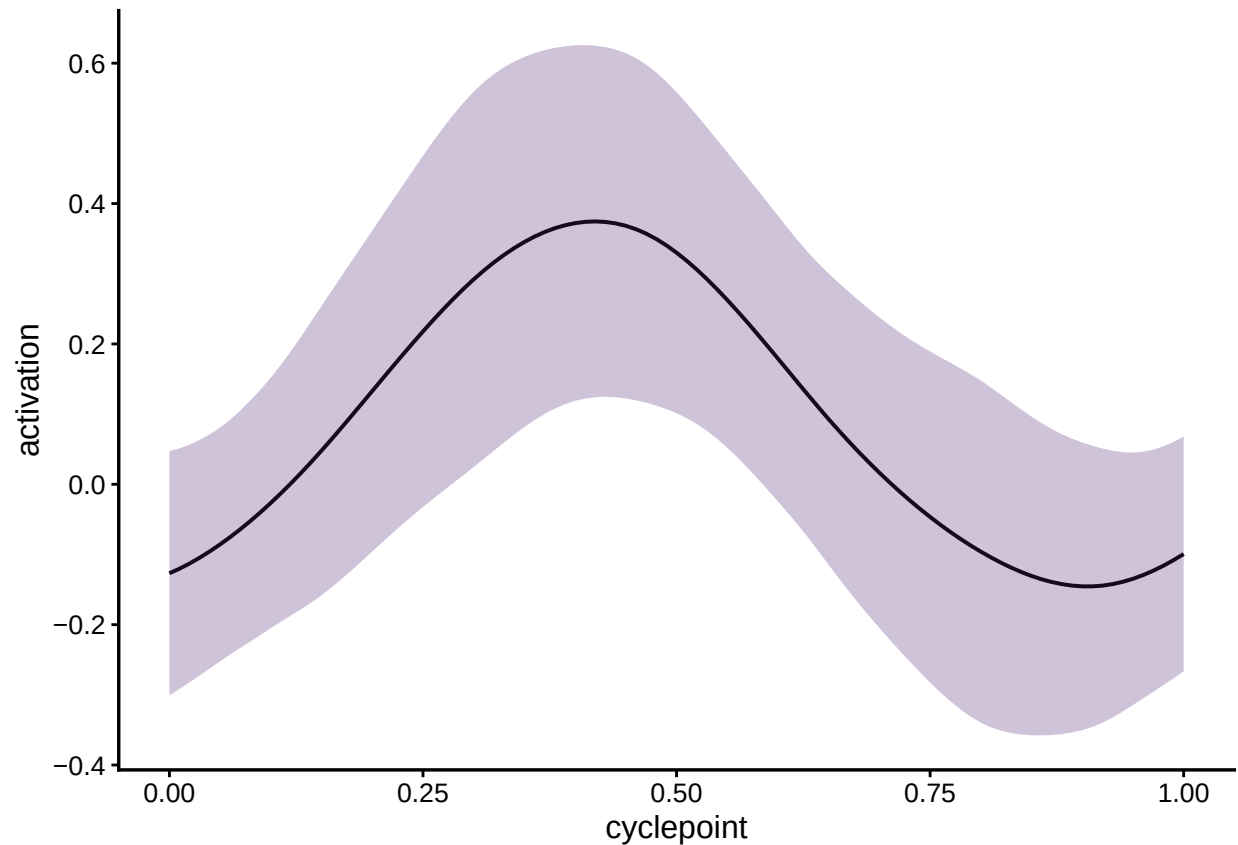
```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## parcel_99 ~ s(cyclepoint, bs = "cc")
##
## Parametric coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.06155    0.09375   0.657   0.515
##
## Approximate significance of smooth terms:
##               edf Ref.df    F p-value
## s(cyclepoint) 0.8001     8 0.228  0.243
##
## R-sq.(adj) =  0.0279
## lmer.REML = 80.227  Scale est. = 0.36912  n = 42
```



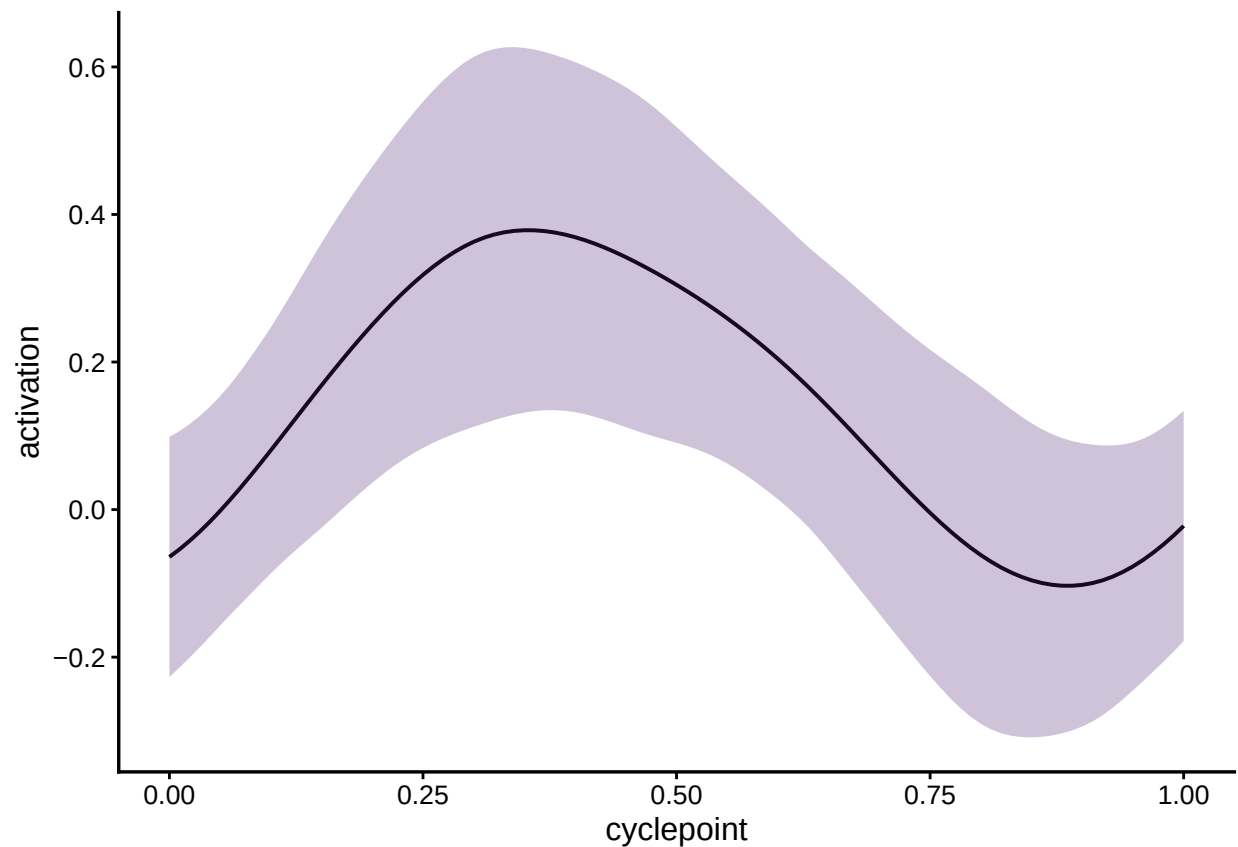
```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## parcel_58 ~ s(cyclepoint, bs = "cc")
##
## Parametric coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.08879    0.07297  -1.217   0.231
##
## Approximate significance of smooth terms:
##              edf Ref.df      F p-value
## s(cyclepoint) 1.55      8 1.505  0.0813 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.0868
## lmer.REML = 61.131  Scale est. = 0.22366  n = 42
```



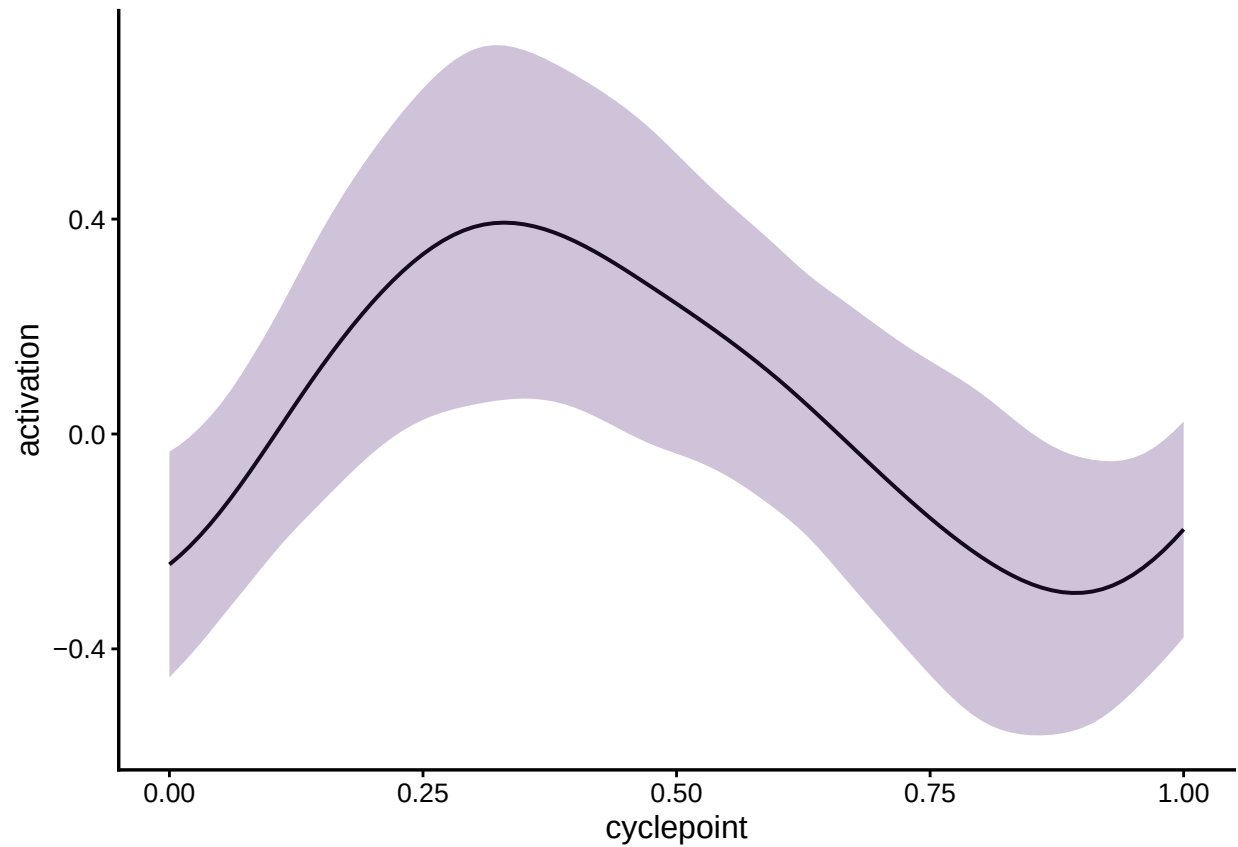
```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## parcel_23 ~ s(cyclepoint, bs = "cc")
##
## Parametric coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.04750    0.05708   0.832    0.41
##
## Approximate significance of smooth terms:
##               edf Ref.df      F p-value
## s(cyclepoint) 2.14      8 8.028 0.00461 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.206
## lmer.REML = 42.566  Scale est. = 0.13684  n = 42
```



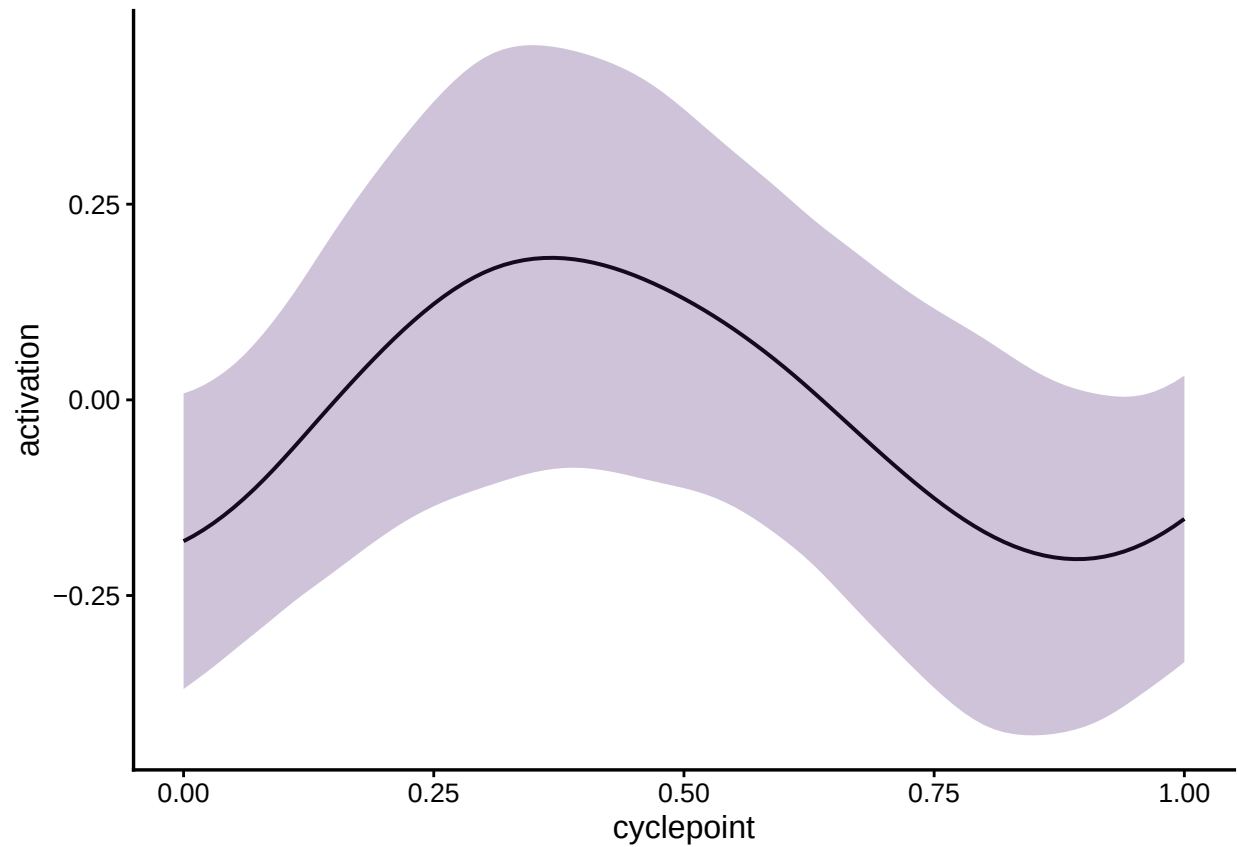
```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## parcel_53 ~ s(cyclepoint, bs = "cc")
##
## Parametric coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.10399    0.05316   1.956  0.0577 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##             edf Ref.df    F p-value
## s(cyclepoint) 2.17     8 8.381 0.00753 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.194
## lmer.REML = 36.818  Scale est. = 0.11868  n = 42
```



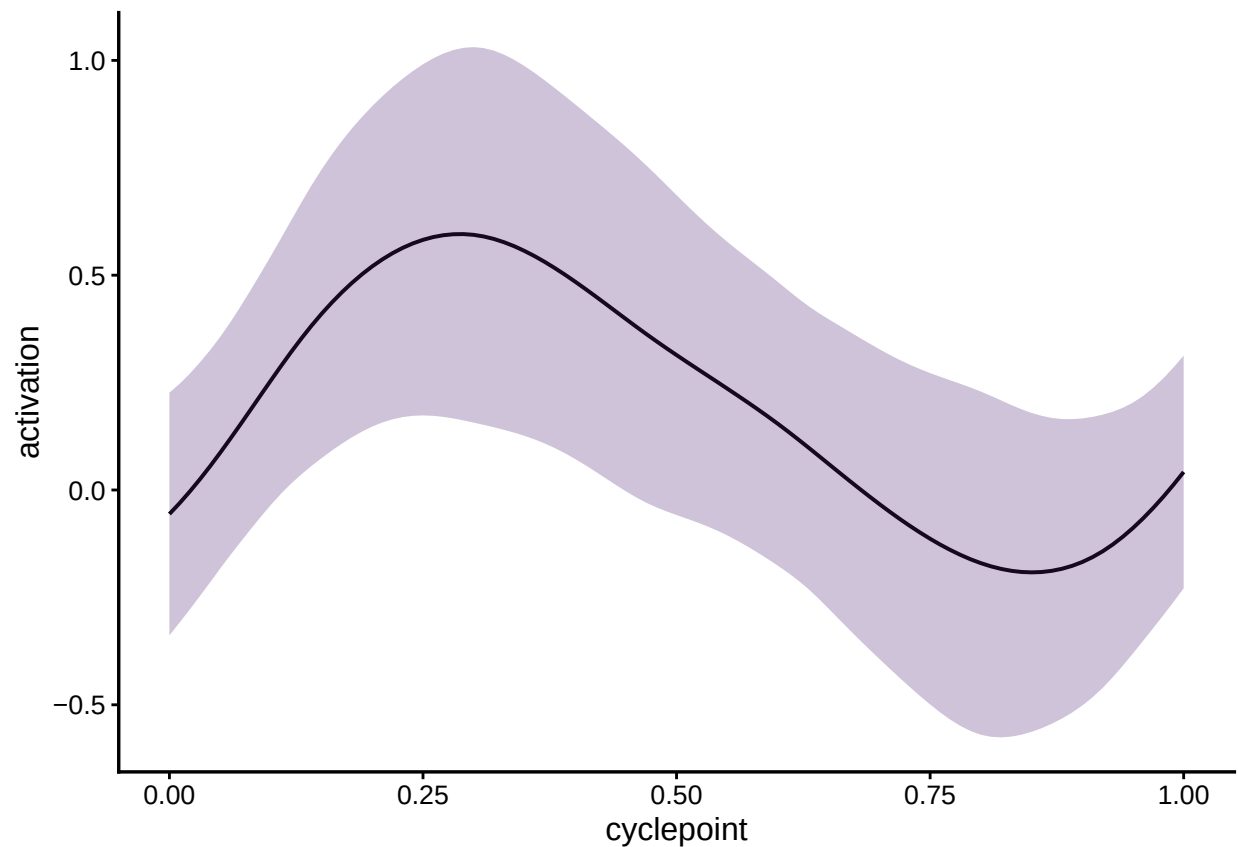
```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## parcel_3 ~ s(cyclepoint, bs = "cc")
##
## Parametric coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.006284  0.067039  -0.094    0.926
##
## Approximate significance of smooth terms:
##              edf Ref.df      F p-value
## s(cyclepoint) 2.371     8 6.428 0.00476 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.221
## lmer.REML = 56.455  Scale est. = 0.18876  n = 42
```



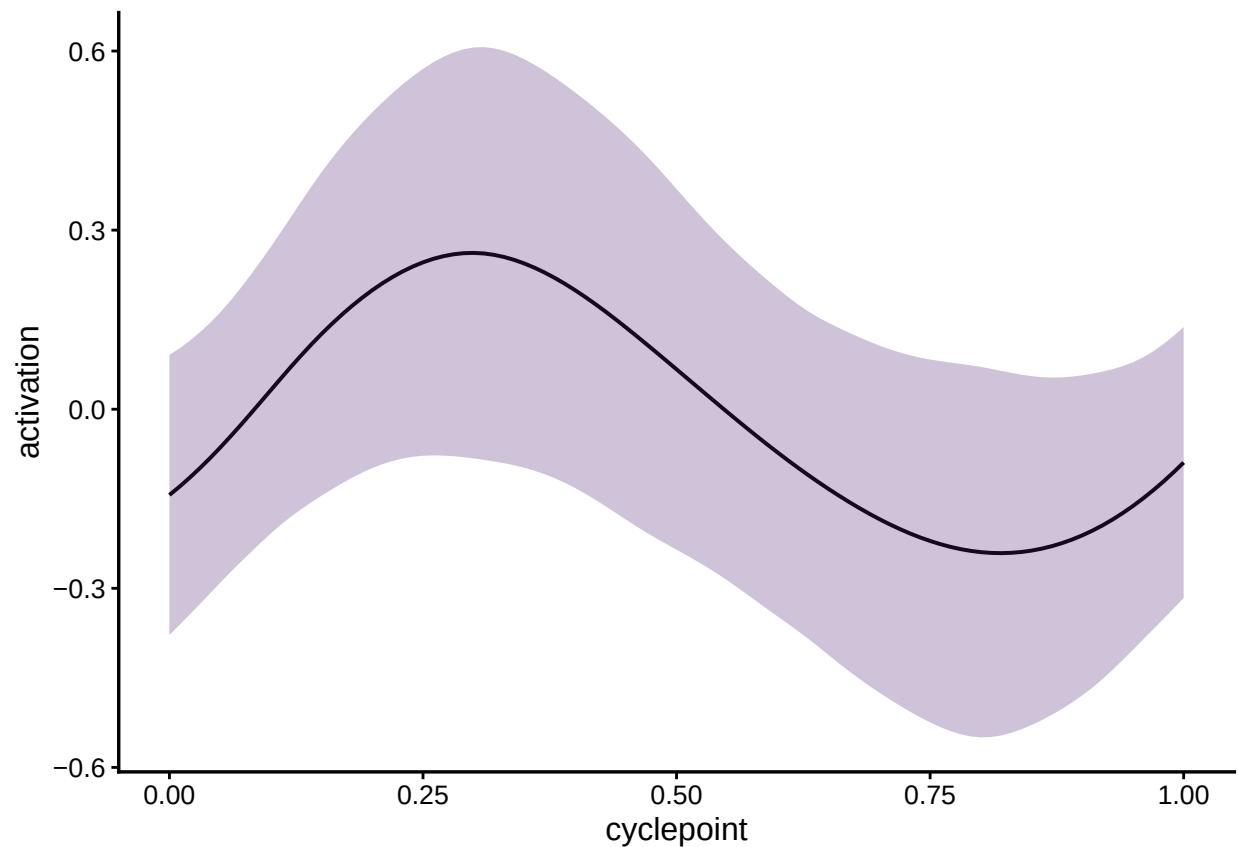
```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## parcel_67 ~ s(cyclepoint, bs = "cc")
##
## Parametric coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.04461    0.06535  -0.683   0.499
##
## Approximate significance of smooth terms:
##             edf Ref.df   F p-value
## s(cyclepoint) 1.712     8 2.6  0.0461 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.111
## lmer.REML = 52.483  Scale est. = 0.17938  n = 42
```



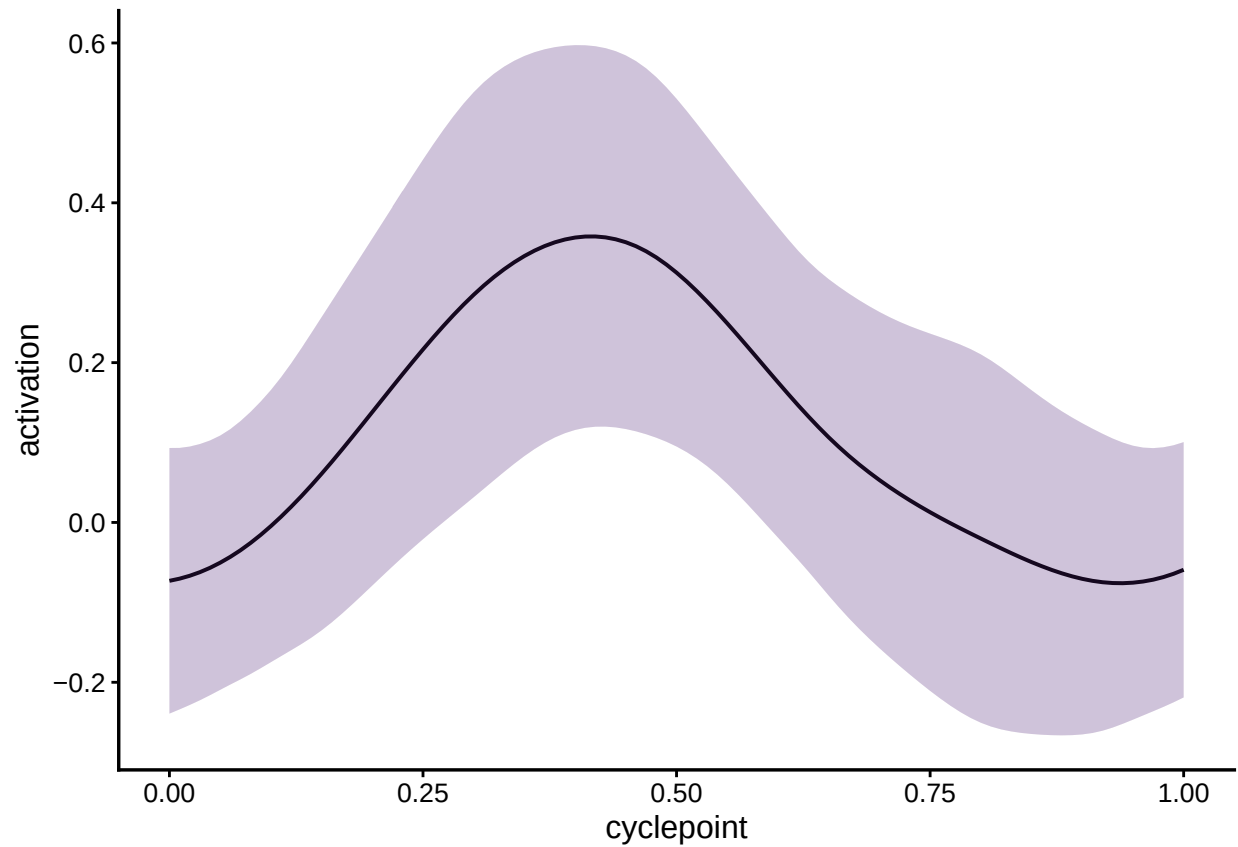
```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## parcel_36 ~ s(cyclepoint, bs = "cc")
##
## Parametric coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.14927    0.09187   1.625   0.112
##
## Approximate significance of smooth terms:
##             edf Ref.df      F p-value
## s(cyclepoint) 2.207     8 2.189 0.0269 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.157
## lmer.REML = 81.795  Scale est. = 0.35451  n = 42
```

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## parcel_49 ~ s(cyclepoint, bs = "cc")
##
## Parametric coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.03725    0.08042  -0.463   0.646
##
## Approximate significance of smooth terms:
##              edf Ref.df      F p-value
## s(cyclepoint) 1.783     8 1.559  0.0754 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.103
## lmer.REML = 69.684  Scale est. = 0.27166  n = 42
```



```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## parcel_30 ~ s(cyclepoint, bs = "cc")
##
## Parametric coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.07129    0.05486   1.299   0.201
##
## Approximate significance of smooth terms:
##              edf Ref.df      F p-value
## s(cyclepoint) 2.083     8 6.746 0.0121 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.174
## lmer.REML = 39.142  Scale est. = 0.12639  n = 42
```



```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## parcel_56 ~ s(cyclepoint, bs = "cc")
##
## Parametric coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.03921    0.05857  -0.669   0.507
##
## Approximate significance of smooth terms:
##              edf Ref.df    F p-value
## s(cyclepoint) 1.978     8 4.613 0.0251 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.146
## lmer.REML = 44.215  Scale est. = 0.14409  n = 42
```

